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TASK 1: POLYMORPHISM OVERLOADING ADDITION METHODS

```
1 class Shape {  
2  
3     void area() {  
4         System.out.println("Area of shape is undefined");  
5     }  
6 }  
7  
8 class Circle extends Shape {  
9  
10     double radius = 5;  
11  
12     @Override  
13     void area() {  
14         double result = 3.14 * radius * radius;  
15         System.out.println("Area of Circle: " + result);  
16     }  
17 }  
18  
19 class Cylinder extends Shape {  
20  
21     double radius = 3;  
22     double height = 10;  
23  
24     @Override  
25     void area() {  
26         double result = 2 * 3.14 * radius * (radius + height);  
27         System.out.println("Area of Cylinder: " + result);  
28     }  
29 }  
30  
31 public class Main {  
32  
33     public static void main(String[] args) {
```

```
Area of Circle: 78.5  
Area of Cylinder: 244.92  
Area of shape is undefined  
=== Code Execution Success
```

TASK 2: POLYMORPHISM OVERRIDING, SHAPE CLASS, CYLINDER CLASS, CIRCLE CLASS

Main.java

Share

Run

```
1 class Shape {
2
3     void area() {
4         System.out.println("Area of shape is undefined");
5     }
6 }
7
8 class Circle extends Shape {
9
10     double radius = 5;
11
12     @Override
13     void area() {
14         double result = 3.14 * radius * radius;
15         System.out.println("Area of Circle: " + result);
16     }
17 }
18
19 class Cylinder extends Shape {
20
21     double radius = 3;
22     double height = 10;
23
24     @Override
25     void area() {
```

Output

```
Area of Circle: 78.5
Area of Cylinder: 244.92
Area of shape is undefined

=== Code Execution Successful ===
```

TASK 3: WRITE A JAVA PROGRAM TO CREATE A BASE CLASS BANK ACCOUNT WITH METHODS DEPOSIT () AND WITHDRAW (). METHOD IN EACH SUB CLASS TO IMPOSE DIFFERENT WITHDRAWAL LIMITS AND FEES

```
1 class BankAccount {
2
3     double balance = 10000;
4
5     void deposit(double amount) {
6         balance += amount;
7         System.out.println("Deposited: " + amount);
8         System.out.println("Current Balance: " + balance);
9     }
10
11    void withdraw(double amount) {
12        if (amount <= balance) {
13            balance -= amount;
14            System.out.println("Withdrawn: " + amount);
15            System.out.println("Remaining Balance: " + balance);
16        } else {
17            System.out.println("Insufficient Balance");
18        }
19    }
20 }
21
22 // ----- Savings Account -----
23 class SavingsAccount extends BankAccount {
24
25     @Override
26     void withdraw(double amount) {
27         double fee = 50;
28         if (amount + fee <= balance) {
29             balance -= (amount + fee);
30             System.out.println("Savings Account Withdrawal: " + amount);
31             System.out.println("Fee Charged: " + fee);
32             System.out.println("Remaining Balance: " + balance);
```

----- Savings Account -----
Deposited: 2000.0
Current Balance: 12000.0
Savings Account Withdrawal: 1500.0
Fee Charged: 50.0
Remaining Balance: 10450.0

----- Current Account -----
Deposited: 3000.0
Current Balance: 13000.0
Current Account Withdrawal: 2000.0
Fee Charged: 100.0
Remaining Balance: 10900.0

=== Code Execution Successful ===